

Assignment 5.1: Automatic Garage Door Controller

Ziad Mostafa Abdelaziz

July 22, 2026

1 Introduction

A Finite State Machine (FSM) is a foundational sequential logic tool used to implement control units and sequencers in digital circuits. A standard FSM consists of a finite set of inputs, outputs, and states. At each active clock edge, the next state and output values are computed based on the current state and inputs.

FSMs fall into two architectural categories:

- **Moore FSM:** The outputs are determined solely by the current state variables, meaning outputs change synchronously with the active clock edge.
- **Mealy FSM:** The outputs are a function of both the current state and the current inputs, allowing outputs to change asynchronously as inputs change.

This assignment implements an **Automatic Garage Door Controller** modeled as a Moore FSM to control both the Up and Down motors of a garage door based on user input and limit sensors.

2 Specifications & Block Interface

The controller interfaces with the system via the following ports:

- **CLK (Input):** 50 MHz clock signal driving the synchronous transitions.
- **RST (Input):** Asynchronous active-low reset signal. When active, it initializes the state machine to the IDLE state.
- **Activate (Input):** Push button trigger. It initiates opening (**Mv_Up**) when the door is closed, or closing (**Mv_Dn**) when the door is open.
- **UP_Max (Input):** Sensor signal indicating that the door is completely open.
- **DN_Max (Input):** Sensor signal indicating that the door is completely closed.
- **UP_M (Output):** Enable signal for the Up motor (activates in state **Mv_Up**).
- **DN_M (Output):** Enable signal for the Down motor (activates in state **Mv_Dn**).

3 State Diagram & Transition Logic

The FSM is structured with three unique states:

1. **IDLE**: Default state. Both motors are disabled ($UP_M = 0$, $DN_M = 0$).
2. **Mv_Up**: Up motor enabled ($UP_M = 1$, $DN_M = 0$).
3. **Mv_Dn**: Down motor enabled ($UP_M = 0$, $DN_M = 1$).

3.1 Moore Output Table

State	UP_M	DN_M
IDLE	0	0
Mv_Up	1	0
Mv_Dn	0	1

Table 1: State-dependent Output Assignments (Moore)

3.2 State Transitions

- In **IDLE**, if **Activate** is asserted:
 - Transition to **Mv_Up** if $DN_Max = 1$ (door is fully closed).
 - Transition to **Mv_Dn** if $UP_Max = 1$ (door is fully open).
 - Remain in **IDLE** otherwise (including intermediate positions where both sensors are 0).
- In **Mv_Up**, remain in state until the top sensor $UP_Max = 1$ is reached, which triggers a transition back to **IDLE**.
- In **Mv_Dn**, remain in state until the bottom sensor $DN_Max = 1$ is reached, which triggers a transition back to **IDLE**.

4 Simulation Waveforms & Analysis

The functional behavior of the controller was validated in ModelSim using a 50 MHz clock (period = 20 ns).

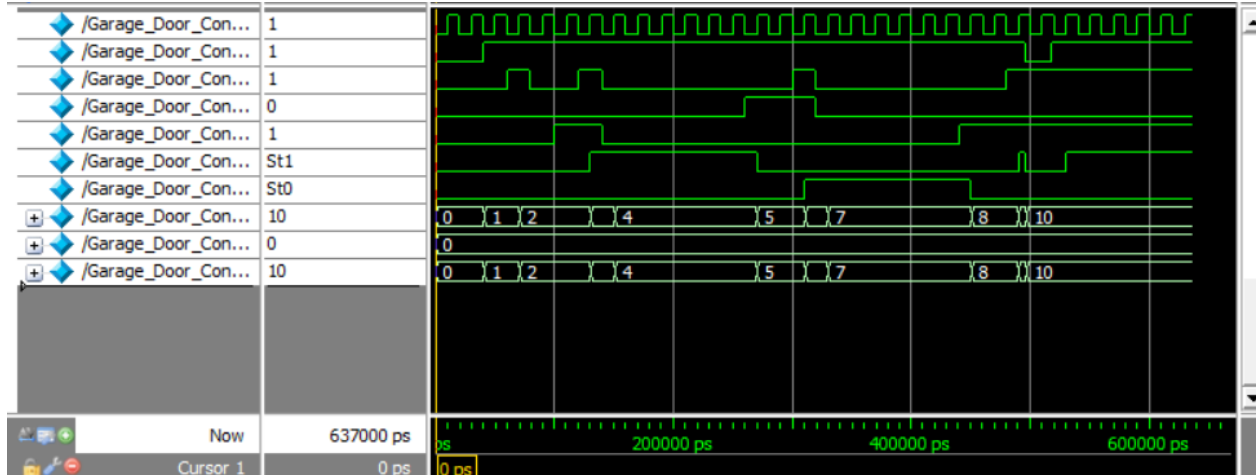


Figure 1: Functional Simulation Waveforms of the Garage Door Controller

4.1 Detailed Cycle-by-Cycle Analysis

As displayed in Figure 1, the state transitions and outputs behave as follows:

- **Asynchronous Reset (Time 0 to 40 ns):** During this startup phase, the reset line `RST` is held low. The state machine resides in the `IDLE` state, with both motor outputs `UP_M` and `DN_M` forced to 0.
- **Activate at Intermediate Position (Time 40 ns to 80 ns):** At 40 ns, `RST` is released. At the negative clock edge around 50 ns, `Activate` is pulsed high while both sensor inputs (`UP_Max` and `DN_Max`) remain at 0 (representing a partially open door). On the following rising clock edge (60 ns), the FSM correctly remains in `IDLE`, keeping both motor outputs low.
- **Door Opening Cycle (Time 90 ns to 270 ns):**
 - At 90 ns, the door-closed sensor `DN_Max` rises to 1.
 - At 110 ns, the user pulses `Activate` = 1. On the next rising clock edge at 120 ns, the FSM transitions to the `Mv_Up` state, causing the output `UP_M` to assert high.
 - At 150 ns, `Activate` returns to 0 and the door leaves the bottom limit switch (`DN_Max` = 0). The FSM remains in `Mv_Up` and `UP_M` stays high, showing the system holds state self-sufficiently.
 - At 250 ns, the top limit switch `UP_Max` goes high. On the rising edge at 260 ns, the FSM transitions back to `IDLE`, disabling the Up motor (`UP_M` = 0).
- **Door Closing Cycle (Time 290 ns to 450 ns):**
 - The door remains open at the top with `UP_Max` = 1.
 - At 310 ns, `Activate` is asserted. On the rising edge at 320 ns, the state transitions to `Mv_Dn`, asserting the Down motor (`DN_M` = 1).
 - At 350 ns, `Activate` is deasserted and `UP_Max` falls to 0. The motor output `DN_M` remains active.

- At 430 ns, the door hits the bottom limit switch (DN_Max = 1). On the rising edge at 440 ns, the FSM returns to IDLE, disabling the Down motor (DN_M = 0).
- **Asynchronous Interruption Verification (Time 470 ns to 500 ns):** To verify the asynchronous property of the reset, the opening sequence is re-initiated. **Activate** is pulsed at 490 ns, starting the Up motor on the clock edge. In the middle of the cycle, at exactly 496 ns, the reset line **RST** is pulled low. The motor output **UP_M** immediately drops to 0 within less than 1 ns, verifying that the reset behaves asynchronously and overrides clock transitions.

5 Simulation Testbench Log

The testbench validates all operational phases using automated checking assertions, resulting in a successful test execution report:

```

# transcript
# [191000 ns] TEST 4 PASSED : UP_M stays active when rising (DN_Max leaves) | UP_M=1, DN_M=0
# [271000 ns] TEST 5 PASSED : Reaching UP_Max clears UP_M (back to IDLE) | UP_M=0, DN_M=0
# [311000 ns] TEST 6 PASSED : Start Moving Down (DN_M=1) when activated and open | UP_M=0, DN_M=1
# [331000 ns] TEST 7 PASSED : DN_M stays active when falling (UP_Max leaves) | UP_M=0, DN_M=1
# [451000 ns] TEST 8 PASSED : Reaching DN_Max clears DN_M (back to IDLE) | UP_M=0, DN_M=0
# [491000 ns] TEST 9 PASSED : Moving Up (Verification before reset check) | UP_M=1, DN_M=0
# [497000 ns] TEST 10 PASSED : Asynchronous reset stops motor immediately | UP_M=0, DN_M=0
# =====
# RESULTS: 10 PASSED, 0 FAILED (out of 10 assertions)
# =====
# ** Note: $finish : E:/courses/Digital IC design Course/ASIC Flow/My Work/My Assignments/Assignment 5.1/src/Garage_Door_Controller_tb.v(i64)
# Time: 637 ns Iteration: 0 Instance: /Garage_Door_Controller_tb
# 1

```

Figure 2: ModelSim Simulation Transcript Output